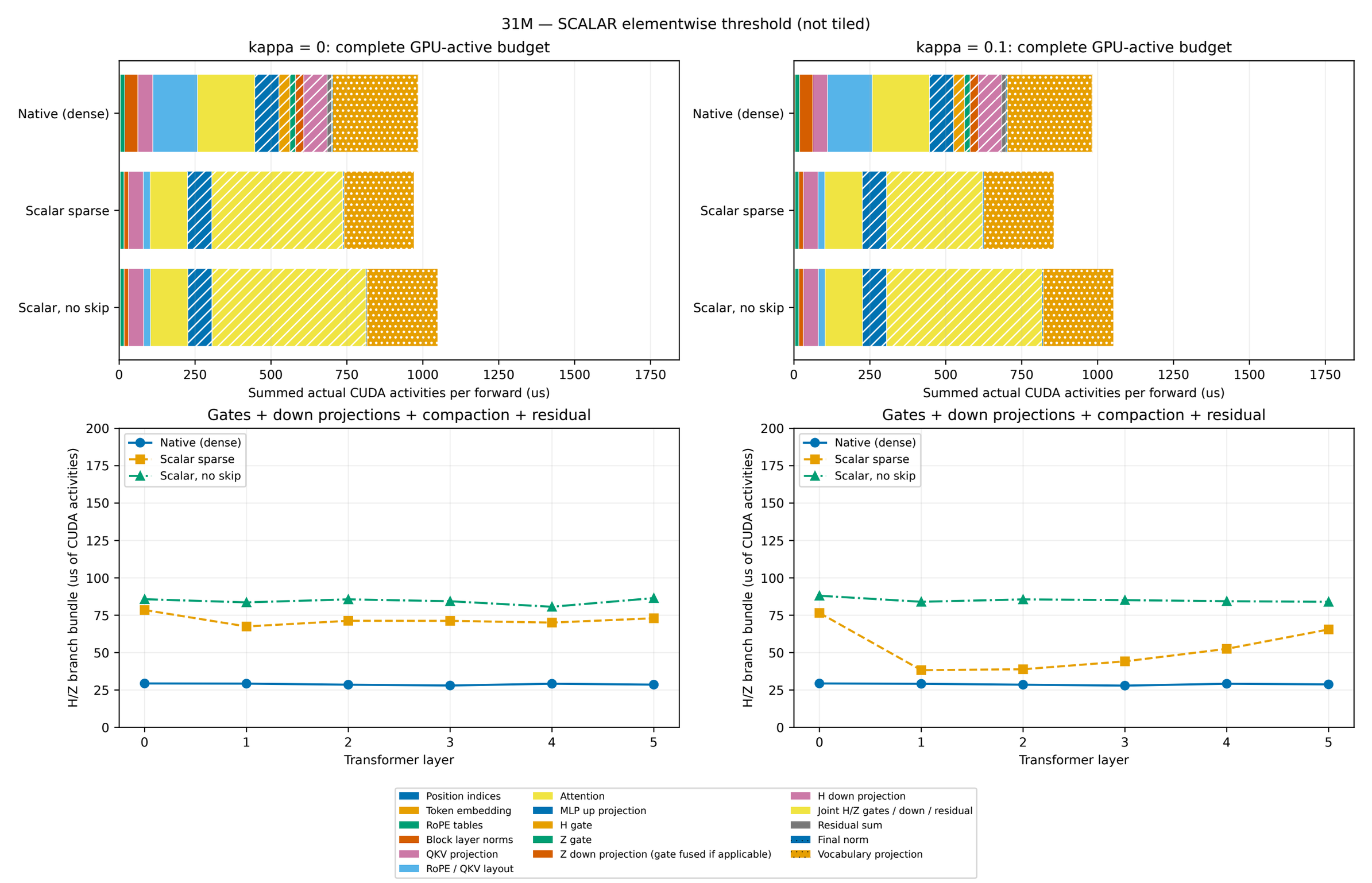

31M — SCALAR elementwise threshold (not tiled)
kappa = 0: complete GPU-active budget
kappa = 0.1: complete GPU-active budget
Native (dense)
Scalar sparse
Scalar, no skip
0
250
500
750
1000
1250
1500
1750
Summed actual CUDA activities per forward (us)
Gates + down projections + compaction + residual
H/Z branch bundle (us of CUDA activities)
Transformer layer
Position indices
Token embedding
RoPE tables
Block layer norms
QKV projection
RoPE / QKV layout
Attention
MLP up projection
H gate
Z gate
Z down projection (gate fused if applicable)
H down projection
Joint H/Z gates / down / residual
Residual sum
Final norm
Vocabulary projection